Advanced Statistics

Regression analysis - introduction to linear models

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Structure of the class

The course content is divided into units, each with a short presentation, a live coding session and a practical
Feel free to ask questions at any point throughout the course
For the coding part we will be using R/RStudio

The units for today

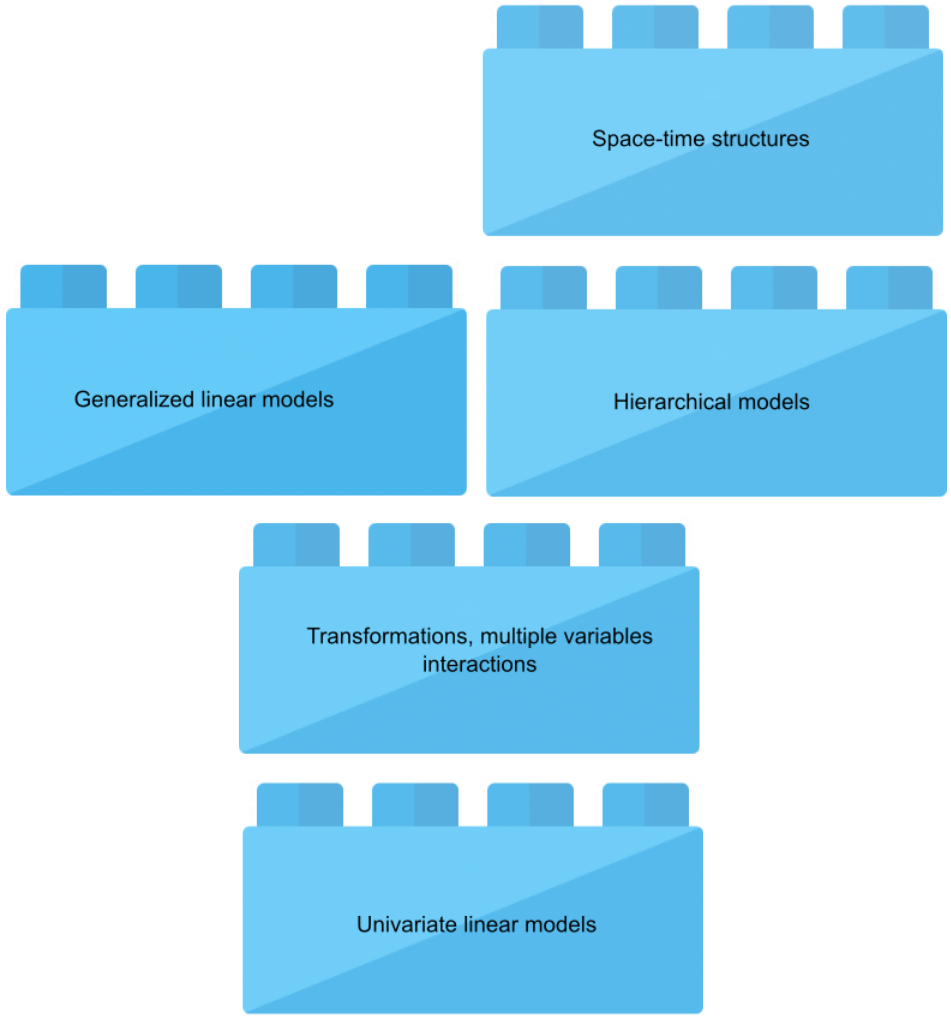
1. regression analysis and introduction to linear models
2. extension of linear models - multiple covariables, transformation, interactions

First a question

Which methods have you already used for data analysis task up until now?

Regression analysis - a definition

- Regression analysis aims to explain and/or predict how a response variable changes as a function of one or more explanatory variables while accounting for uncertainty.
- This framework offer flexibility to fit rich models
- We can add various components to the model as building blocks
- $y_i = f(X_i) + f(S_i) + f(t_i)$



Why are regression analysis still widely used?

Regression remains one of the most useful tools for data analysis in environmental science.

Advantages

- Fast to fit (milliseconds)
- Works well with small to medium sized datasets
- Easy to interpret
- Quantifies effect sizes
- Statistical inference
- Easy to communicate
- Strong theoretical foundations

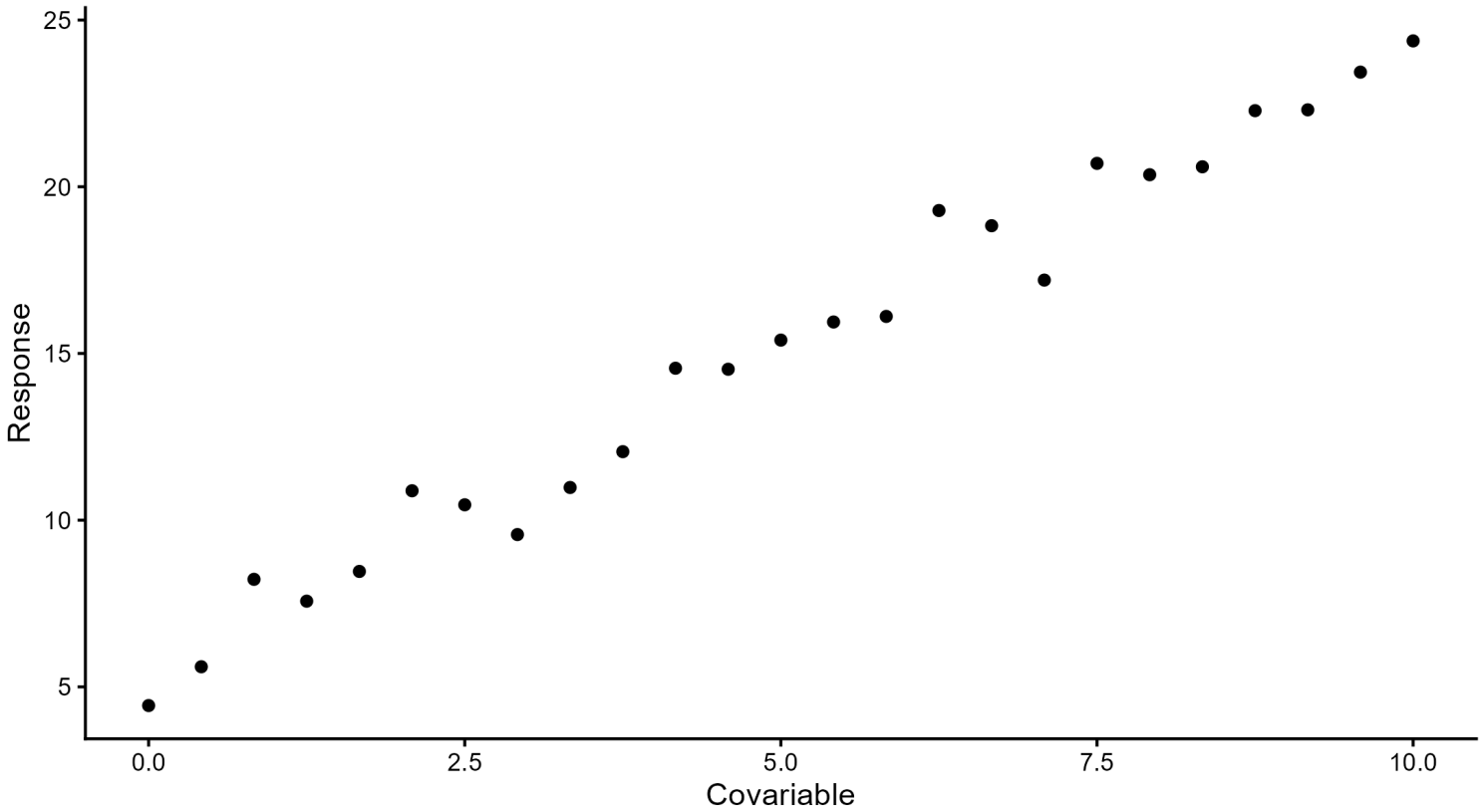
Limitations

- Parametric approach - need to check model assumptions
- Limited for complex interactions
- Cannot automatically learn nonlinear patterns
- May underperform for purely predictive tasks

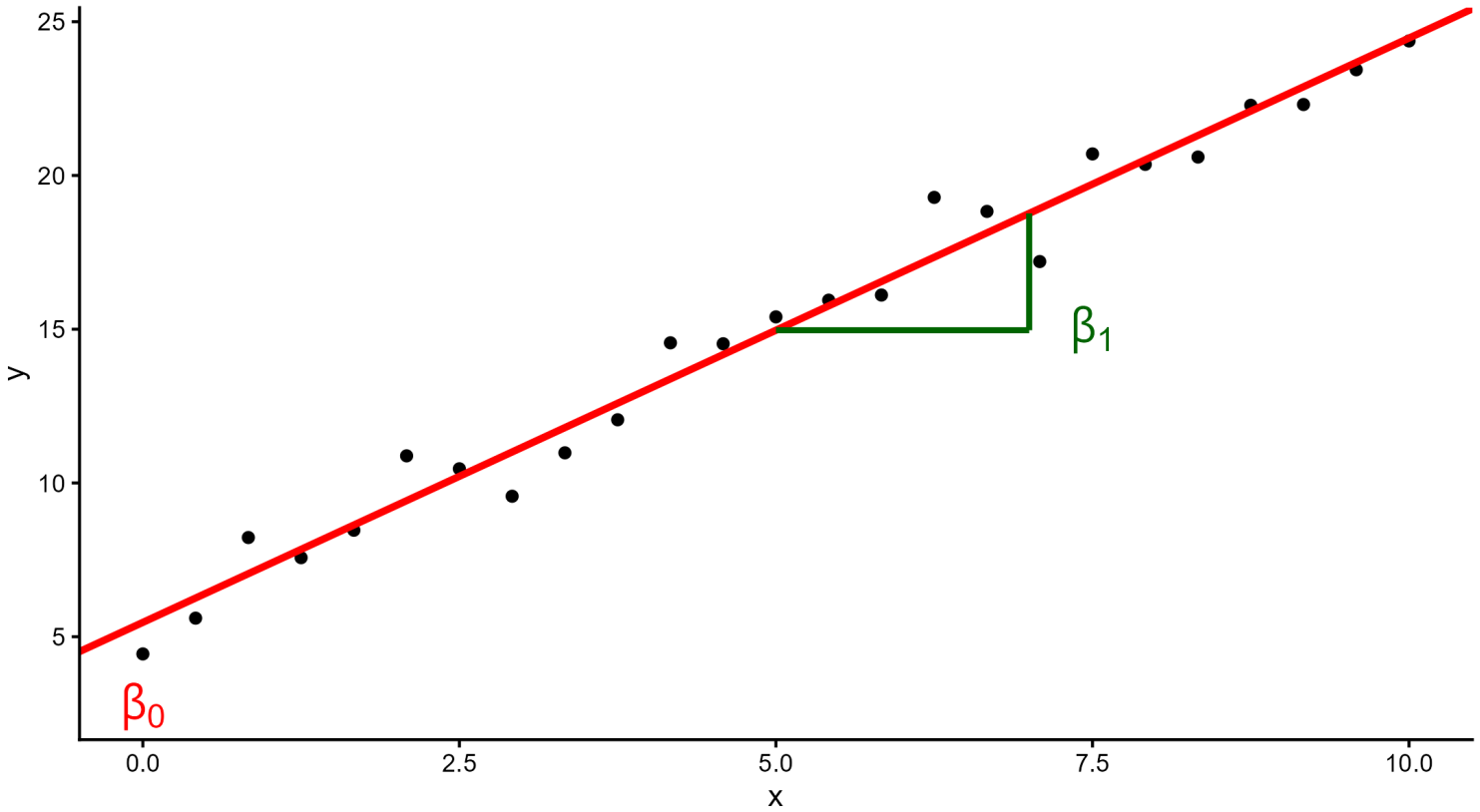
The different steps of a regression analysis

1. Data preparation and formatting (+++)
2. Model estimation (+)
3. Model checks (++)
4. Model interpretation (++)

The base: univariate linear models in 2 figures



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The univariate linear model in one equation

$$response_i = \beta_0 + \beta_1 * covariable_i + \epsilon_i \quad (1)$$

- i is an index for the observation (1, 2, ...)
- β_0 is the intercept, the expected response value when $covariable = 0$
- β_1 is the slope, the unit change in the response per unit change in the covariable

The univariate linear model in one equation

What is the ϵ term?

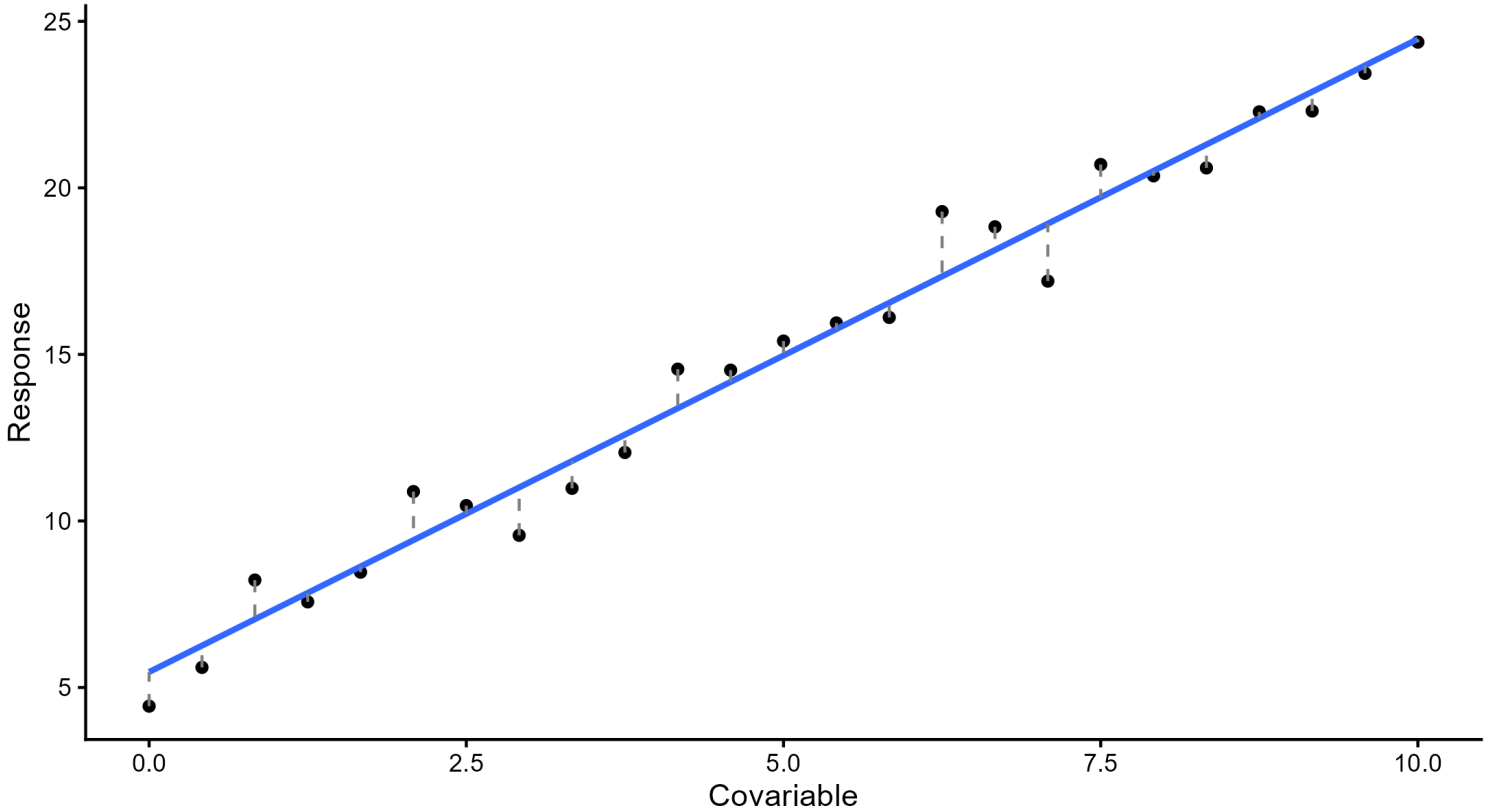
$$response_i = \beta_0 + \beta_1 * covariable_i + \epsilon_i \quad (2)$$

Options:

- The uncertainty of the β_0 et β_1 coefficients
- The differences between the regression line and the observations
- The linearity coefficient
- D, la réponse D

The univariate linear model in one equation

Answer: The differences between the regression line and the observations - also called *residuals*



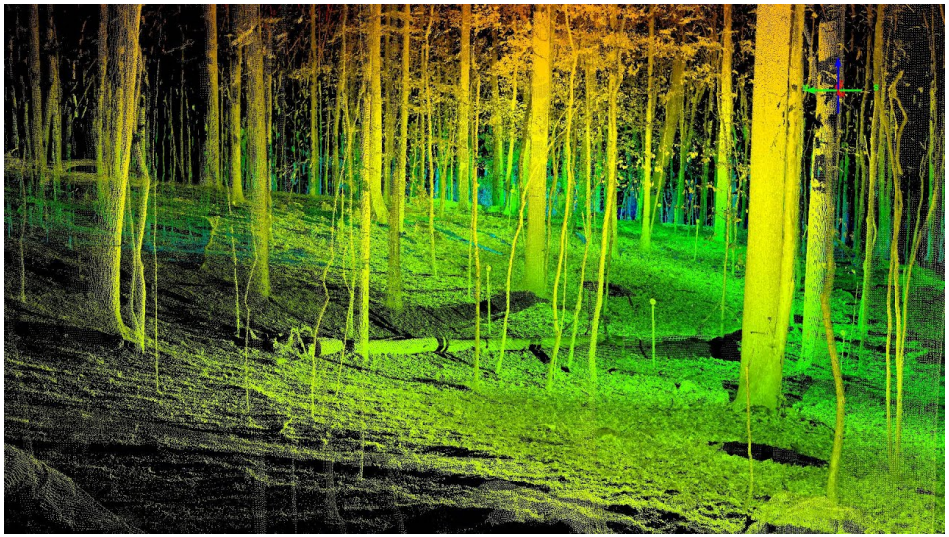
The linear model gives three properties to the residuals

1. normal distribution
2. a mean of 0
3. a standard deviation defined by σ

Checking Model Assumptions

Before interpreting the results of a linear model, it is essential to verify that its assumptions are not violated.
The following should be checked (in decreasing order of importance):

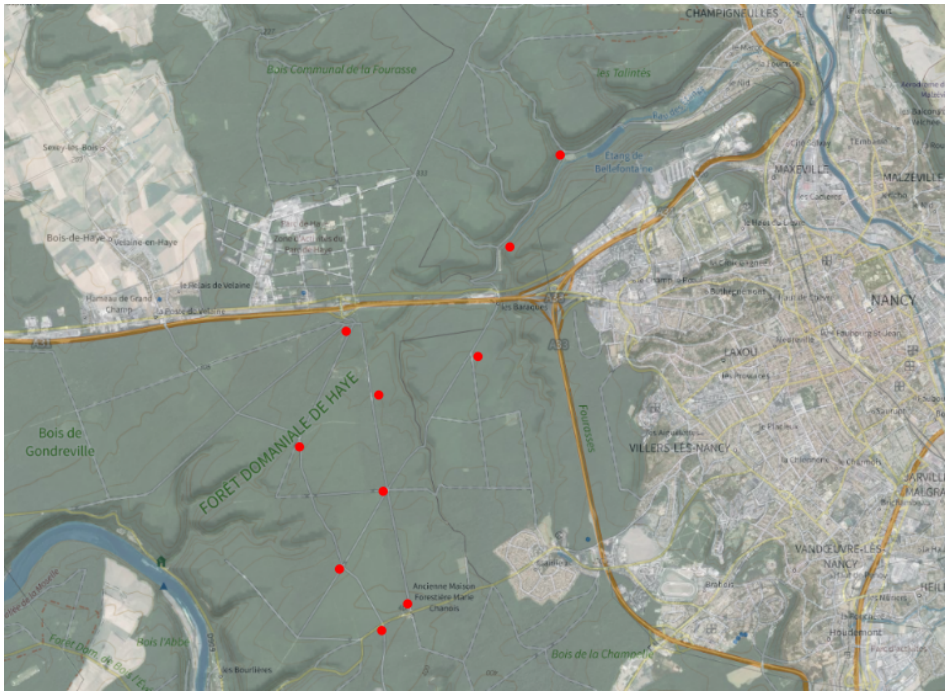
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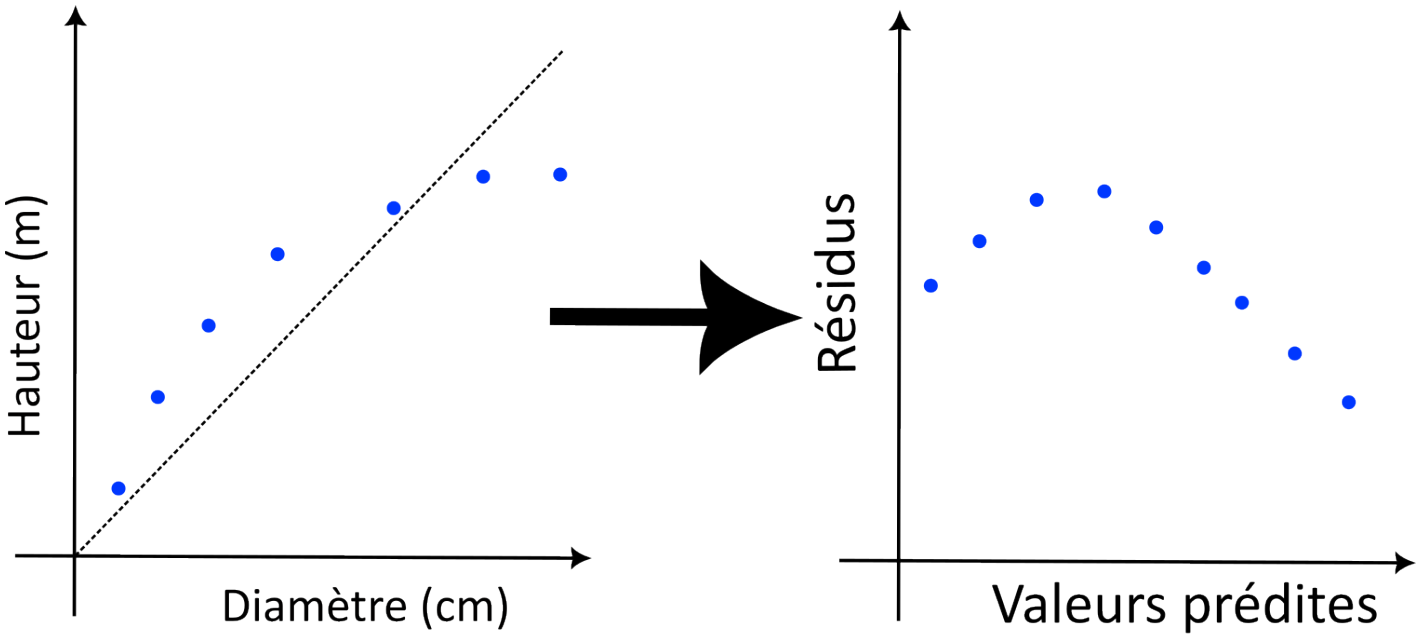
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5. Constant variance of residuals (homoscedasticity)

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5. Constant variance of residuals (homoscedasticity)
6. Normality of residuals

Step 2: Checking Model Assumptions

Linear regression assumes that the relationship between the covariables (x) and the response (y) is linear.
If this assumption is violated, the interpretation of the model coefficients is no longer valid.
Linearity can be assessed by plotting the residuals against the fitted values.



Checking Model Assumptions

Linear models can still be used for some nonlinear relationships by modifying the model, for example:

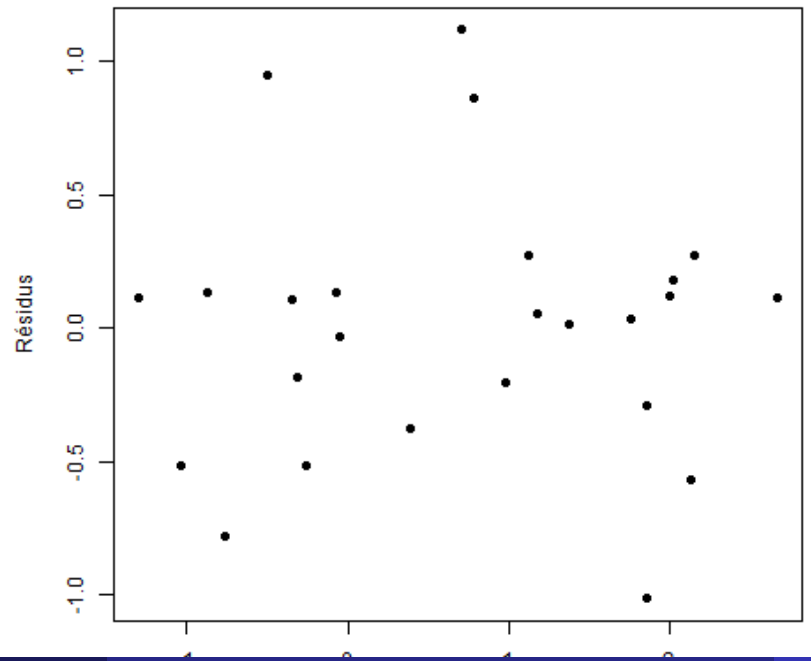
- Polynomial terms
- Transformations of the predictor(s) (log, square root, ...)
- Transformations of the response (log, square root, ...)

We will look into this in more details in the next unit.

Checking Model Assumptions

Linear regression assumes that the residual variance is constant (homoscedasticity).
This assumption is assessed by plotting residuals against fitted values.

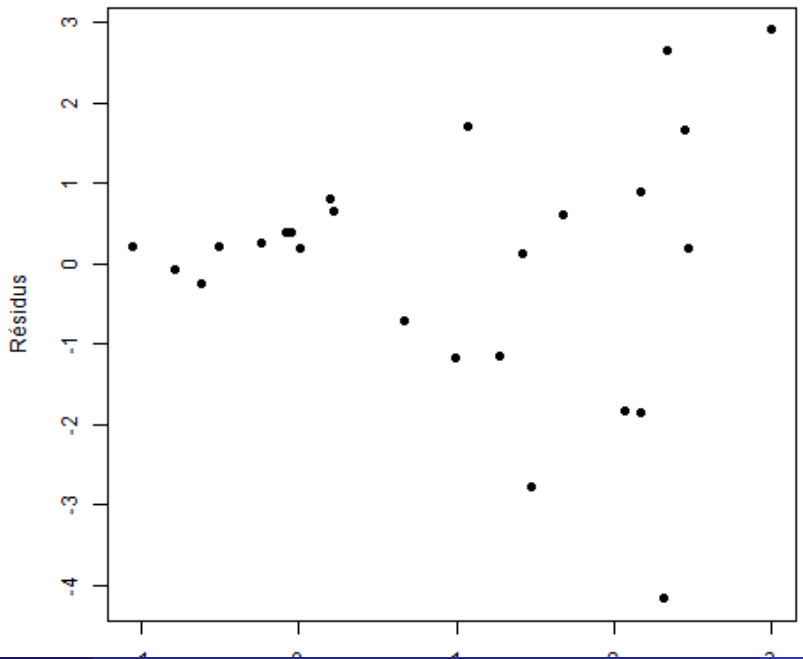
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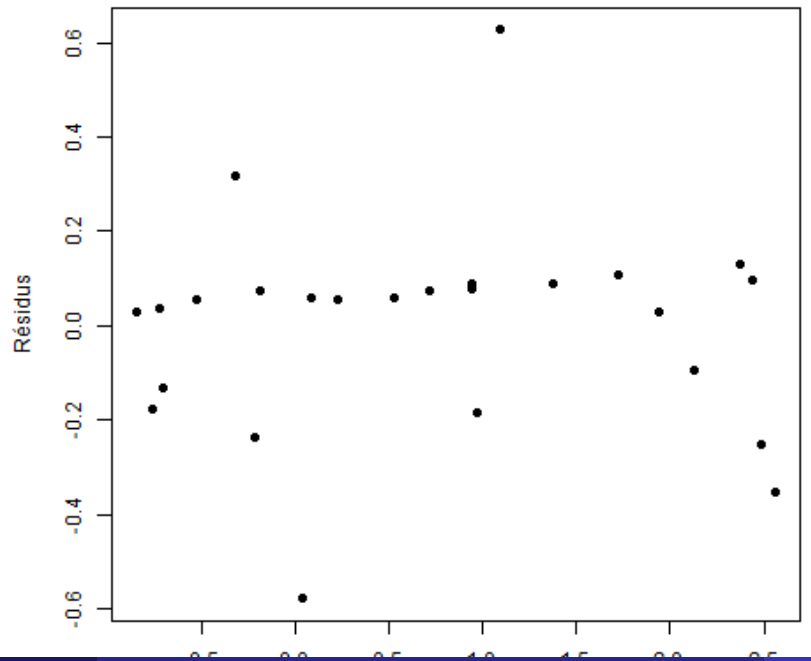
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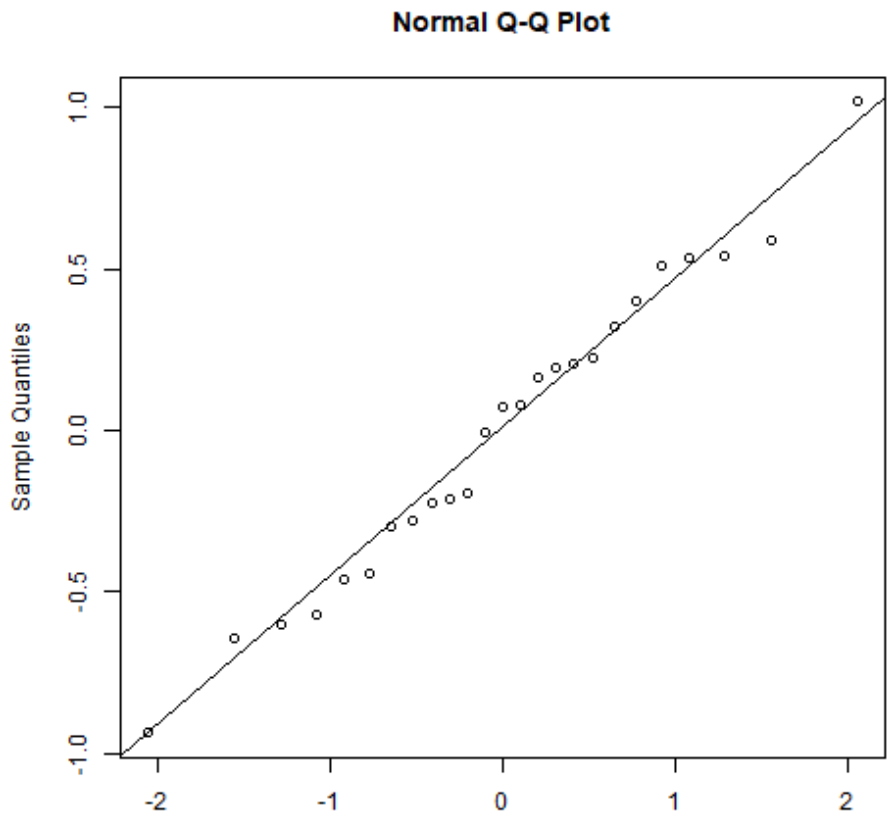
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Checking Model Assumptions

The residuals of a linear model should approximately follow a normal distribution.
This assumption can be assessed using a Q–Q plot, which compares the observed residual quantiles with the theoretical quantiles of a normal distribution.



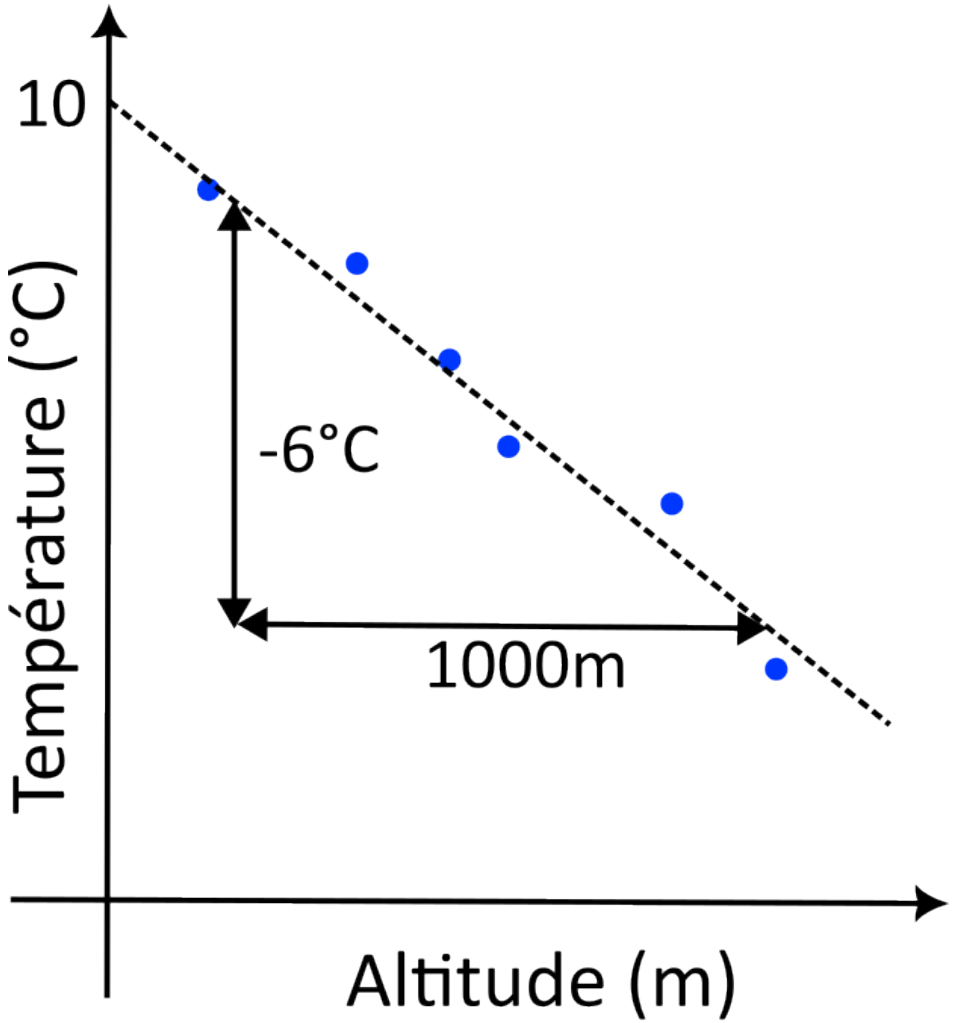
Model Interpretation

Suppose we have this model:

$$\text{Temperature } (^\circ\text{C}) = 10 - 0.006 \times \text{Altitude } (\text{m}) + \varepsilon$$

This model can be interpreted as follows:

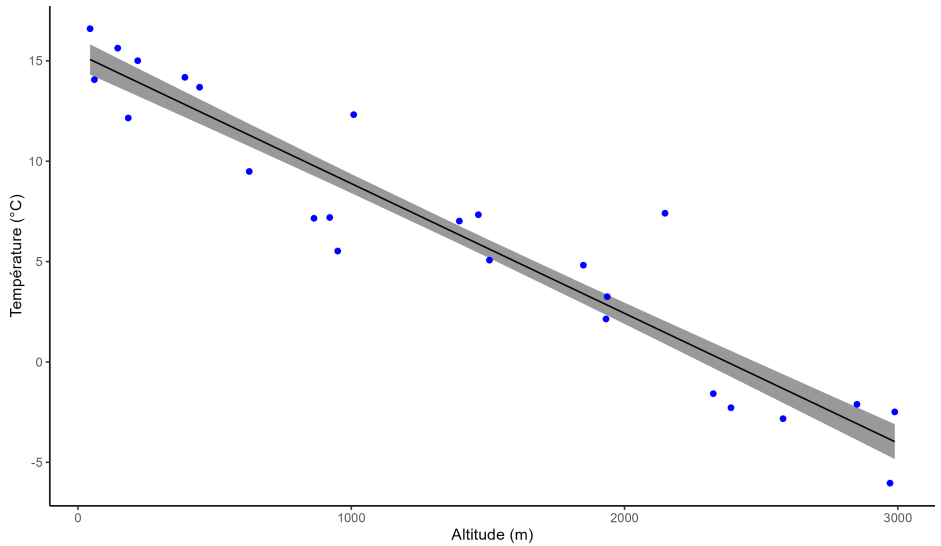
- At an altitude of 0 m, the expected temperature is 10°C.
- Two locations differing by 1000 m in altitude are expected to differ by 6°C on average.
- At an altitude of -500 m, the predicted temperature is 13°C.



Model Interpretation

Linear regression also quantifies the uncertainty associated with the model.

- Uncertainty in the regression coefficients, allowing confidence intervals and hypothesis tests (p-values).
- Uncertainty in the fitted values, allowing confidence intervals around the regression line.



Time for some coding